D.A.K.S.H. (■■■■)

Decision & Knowledge Synthesizer for Human-operations

System Architecture, Branching Strategy & Engineering Implementation Plan

1. EXECUTIVE ARCHITECTURAL VISION

D.A.K.S.H. is an autonomous cognitive personal operating system designed to transcend ordinary task list and assistant applications. Combining deep Android OS hardware authority, real-time ambient notification interception, alternate-day mathematical routine scheduling, and a low-latency cinematic baritone voice loop, D.A.K.S.H. serves as a continuous sovereign life copilot. The design language rejects typical AI clichés in favor of a **Zerodha Kite-inspired** high-density, tactical light-default interface paired with an obsidian dark theme.

2. BRANCH & SUB-BRANCH IMPLEMENTATION ROADMAP

The engineering lifecycle is structured across five primary branches and eleven focused sub-branches:

BRANCH IDENTIFIER	SUB-BRANCHES & DOMAIN	KEY DELIVERABLES
`arch/core-engine`	• 1.1: Event Bus & Reactive State • 1.2: Encrypted Local Store • 1.3: Health Sentinel	Thread-safe pub/sub dispatch bus, local encrypted SQLite, battery and latency metrics tracker.
`agent/cognitive-brain`	• 2.1: Tactical Router • 2.2: Temporal & Routine Matrix • 2.3: Device Action Dispatcher • 2.4: Persona Engine	<80ms intent classification, modulo alternate-day engine, midnight auto-rollover, contact updater, baritone persona.
`platform/android-sentinel`	• 3.1: Consent & Auth Protocol • 3.2: Contacts Controller • 3.3: Notification Interceptor • 3.4: Doze Persistence	NotificationListenerService, ContactsContract CRUD, exact hardware alarms, and background foreground service.
`voice/neural-acoustic`	• 4.1: Streaming ASR • 4.2: Baritone Neural Voice • 4.3: Realtime Duplex	Fast wake-word detection, studio-weighted cinematic voice (Vision/Jarvis cadence), instant barge-in interruption.
`ui/kite-tactical`	• 5.1: Dual Theme (Light Default) • 5.2: High-Density HUD • 5.3: Harmonic Spectrum Meter	Kite-inspired financial-grade ledger, paper-white default UI, obsidian dark mode, live audio reactive meter.

3. SOVEREIGN MOBILE CONTROL & CONSENT PROTOCOL

D.A.K.S.H. executes system operations with verified user consent. Unlike cloud-only assistants, it bridges directly to native Android subsystems:

SYSTEM DOMAIN	ANDROID PERMISSION / API	D.A.K.S.H. CAPABILITY & VOICE ACTION

`Address Book`	READ_CONTACTS WRITE_CONTACTS	Allows spoken mutations: *'Daksh, rename contact Priya to Swimi.'* Queries ContentResolver and updates record in milliseconds.
`Ambient Telemetry`	NotificationListenerService	Triages incoming notifications (WhatsApp, Bank, Delivery), dropping noise and turning actionables into dynamic ledger items.
`Temporal Alerts`	SCHEDULE_EXACT_ALARM WAKE_LOCK	Guarantees morning briefings and alternate-day reminders execute reliably, even when Android enters deep Doze battery mode.

4. KITE-TACTICAL DESIGN SYSTEM (LIGHT & DARK MODES)

The interface rejects standard AI chatbot patterns. It employs an instrument-grade financial HUD:

• **Light Mode (Default):** Crisp paper background (#F8FAFC), 1px slate divider borders (#E2E8F0), dark-slate tabular data (#0F172A), and stark electrical blue accents (#0284C7).
• **Dark Mode (Tactical Obsidian):** Deep OLED black (#0B0D11), matte slate cards (#1E293B), ice-cyan highlights (#00F0FF), and amber rollover markers (#F59E0B).
• **Dynamic Execution Ledger:** Replaces scattered task lists with a 3-column ticket system: Today's Cadences, Rollover Debt Ledger, and Ambient Notification Stream.
• **Harmonic Soundwave Bar:** A continuous, ultra-slim 2px harmonic visualizer that pulses during neural reasoning and voice generation.